Arista Access Node BAS-ACC-06: SOP for Resolving Major Optical Transport Failure (ALARM_8375)

Node: BAS-ACC-06 | **Vendor:** Arista | **Severity:** Major | **Alarm Code:** ALARM_8375

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a major optical transport failure on Arista Access Node BAS-ACC-06 located at Basavanagudi POP. This failure domain is primarily attributed to the degradation of SFP/XFP modules or fiber cuts, resulting in a loss of receive (Rx) or transmit (Tx) power. In Arista networks, the Optical Transport Network (OTN) is responsible for transporting client signals from the access node over fibers. The OTN protocol stack, which includes G.709 and G.7042, ensures efficient transmission of client signals over the fiber infrastructure. However, hardware failures or environmental factors can cause degradation or loss of signal quality, leading to alarm conditions like ALARM_8375. In this context, it is crucial to identify the root cause of the failure and take corrective actions to restore normal operation.

Pre-requisite Checks

- Ensure remote hands are dispatched to Basavanagudi POP for physical access and assistance, if required.

- Verify the status of redundant links and ensure they are functioning as expected.

- Confirm the network configuration and ensure there are no configuration conflicts or misconfigurations that could contribute to the failure.

Potential Root Causes

Cause	Probability
Degraded or faulty SFP/XFP module	High
Fiber cut or damage	Medium
Optical signal quality degradation due to environmental factors (e.g., temperature, humidity)	Low

Diagnostic Commands & Expected Outputs

```
Command: show interfaces | include Optical, Expected Output: Optical signal quality degradation, fiber cut, or SFP/XFP module issues, Description: This command displays the optical signal quality and any issues related to fiber cuts or SFP/XFP module degradation. It helps to identify if the problem is related to the optical signal quality or hardware failure.

Command: show inventory | grep , Expected Output: SFP/XFP module details, including serial number and revision, Description: This command displays the inventory of SFP/XFP modules installed in the device, including the serial number and revision. It helps to identify the specific SFP/XFP module that needs to be replaced or inspected.
```

```
Command: show log | include ALARM_8375, Expected Output: Alarm log entries related to ALARM_8375, Description: This command displays the log entries related to ALARM_8375, which helps to determine the timestamp and duration of the alarm condition.
```

Resolution Steps

Step 1: Step: Replace the faulty SFP/XFP module, Command: install , Description: Replace the faulty SFP/XFP module with a new one. This step requires physical access to the device and the new SFP/XFP module.

Step 2: Step: Inspect and repair or replace the fiber cable, Command: no shutdown , Description: Inspect the fiber cable and repair or replace it if damaged. This step requires physical access to the fiber cable and the new fiber cable, if required.

Post-Check Verification

```
Command: show interfaces | include Optical, Expected Output: Optical signal quality restored to normal, Description: Verify that the optical signal quality has returned to normal after the resolution steps have been taken.

Command: show log | include ALARM_8375, Expected Output: Alarm log entries cleared or no longer related to ALARM_8375, Description: Verify that the alarm log entries related to ALARM_8375 have been cleared or are no longer present after the resolution steps have been taken.
```

Roll-back Procedures

- Step: Restore the original SFP/XFP module, Command: uninstall , Description: Restore the original SFP/XFP module if the replacement caused issues.

- Step: Restore the original fiber cable, Command: shutdown , Description: Restore the original fiber cable if the replacement caused issues.